
BEYOND FLICKERING: INTRODUCING CODE-MODULATED MOTION VISUAL EVOKED POTENTIALS FOR BRAIN-COMPUTER INTERFACING

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Hanneke A. Scheppink*

Faculty of Technology and Bionics
Rhine-Waal University of Applied Sciences
Kleve, Germany

Rainer Herpers*

Institute of Visual Computing and Graduate Institute
Bonn-Rhein-Sieg University of Applied Science
Sankt Augustin, Germany

Jordy Thielen

Radboud University
Donders Institute for Brain, Cognition and Behaviour
Nijmegen, The Netherlands

Ivan Volosyak*

Faculty of Technology and Bionics
Rhine-Waal University of Applied Sciences
Kleve, Germany
ivan.volosyak@hochschule-rhein-waal.de

ABSTRACT

A code-modulated motion visual evoked potential (c-MVEP) for brain-computer interfacing (BCI) is presented in this study. This paradigm uses pseudo-random sequences to visually stimulate objects using motion as an alternative to flickering. In an offline experiment of this study, EEG data were recorded and compared during sequential stimulation of a single object under four conditions: c-MVEP, code-modulated visual evoked potential (c-VEP), steady-state motion visual evoked potential (SSMVEP), and steady-state visual evoked potential (SSVEP). The c-MVEP showed similar time-domain characteristics as the c-VEP, and also in the frequency domain the c-MVEP evoked a broadband response similar to the c-VEP, with a comparable signal-to-noise ratio (SNR), albeit more focused in the lower frequency range. Both the SSMVEP and SSVEP showed clear oscillatory responses at the stimulation frequency and harmonics, with a higher SNR for SSVEP than SSMVEP. The spatial distribution of c-MVEP showed the main activation at Oz and spread across multiple electrodes, whereas c-VEP showed less spreading and was more focused at Oz. Similar observations were made for SSMVEP and SSVEP. From subjective ratings, there was no clear preference for the motion-based stimulation of SSMVEP or c-MVEP over flicker-based stimulation of SSVEP or c-VEP. In an online experiment of this study, we evaluated a 4-class BCI using the same four conditions, testing the practical feasibility of the c-MVEP paradigm. The c-MVEP BCI reached a mean accuracy of 85.67 % with an average selection time of 2.61 s, which was significantly lower than c-VEP (97.81 %; 1.15 s) and SSVEP (93.42 %; 1.94 s), but significantly higher than SSMVEP (64.91 %; 4.18 s). Overall, this study shows the great potential of the newly proposed c-MVEP paradigm using

*Department of Informatics and Data Science, Doctoral School for Applied Research in North Rhine-Westphalia, Bochum, Germany

motion stimulation for BCI applications, providing a valuable alternative to the c-VEP paradigm using flickering stimulation.

Keywords brain-computer interface (BCI) · code-modulated motion visual evoked potential (c-MVEP) · code-modulated visual evoked potential (c-VEP) · electroencephalography (EEG) · BCI user comfort · flicker-free VEP · steady-state visual evoked potential (SSVEP) · steady-state motion visual evoked potential (SSMVEP)

1 Introduction

A brain-computer interface (BCI) is a system that allows real-time communication between the human brain and external devices, bypassing muscular involvement (Wolpaw and Wolpaw, 2012). This makes BCI systems a promising tool for users with impaired or absent muscular function. The neural activity is most commonly measured non-invasively using electroencephalography (EEG) (Wolpaw et al., 2002). A common application of BCI is to enable communication using a BCI speller using external stimuli (Rezeika et al., 2018).

One of the visual stimulation paradigms that achieves a high information transfer rate (ITR) is the code-modulated visual evoked potential (c-VEP) (Martínez-Cagigal et al., 2021; Miao et al., 2024). In a c-VEP-based BCI paradigm, stimuli flicker according to a pseudo-random sequence. Often, a binary sequence is used, producing high-contrast flickering between black and white. A similar and high-performing paradigm is the steady-state visual evoked potential (SSVEP) in which stimuli flicker according to a specific and constant frequency (Vialatte et al., 2010).

Although c-VEP and SSVEP are high-performing paradigms, a major limitation of both is the visual and mental fatigue, as well as eye strain, that can result from the high-contrast flickering of the visual stimuli (Xie et al., 2016; Azadi Moghadam and Maleki, 2023).

To address this issue, the steady-state motion visual evoked potential (SSMVEP) paradigm was proposed as an alternative for BCI applications (Xie et al., 2012, 2016). In SSMVEP, stimuli are modulated through continuous movement rather than flickering, according to a specific frequency. This movement can be implemented, for instance, by radially zooming with periodic expansion and contraction of the stimulus size (Chai et al., 2019, 2020), or through dynamic patterns such as expanding and contracting ring-shaped checkerboards (Han et al., 2017; Yan et al., 2017) or Newton rings (Xie et al., 2012, 2016). The motion typically follows a sinusoidal modulation pattern, where the direction of movement reverses twice per cycle, a rate commonly referred to as the flip-frequency or motion-reversal frequency. The neural responses evoked by attending to these motion-based stimuli resemble those of SSVEPs, although with a weaker activation level. SSMVEP evokes responses across more channels than SSVEP, with greater activation in middle temporal areas, whereas SSVEP was more focused in occipital areas (Han et al., 2018b).

Although SSMVEP relies on motion stimulation, it should not be mistaken for the motion-onset VEP (m-VEP), which is based on the anticipation of a single moving stimulus, typically presented in isolation at a constant speed. In contrast, in SSMVEP all stimuli move simultaneously, each at a distinct frequency. The design of SSMVEP allows for shorter stimulation durations and greater flexibility in the number of targets compared to m-VEP, where ITR is limited due to longer stimulation durations and a restricted number of targets (Guo et al., 2008; Li et al., 2017; Ma et al., 2017). Moreover, SSMVEP avoids the direction-specific adaptation effects, commonly arising in m-VEP, that can potentially cause visual fatigue and a reduced signal-to-noise ratio (SNR) (Xie et al., 2012).

Importantly, SSMVEP is particularly well performing under conditions of high visual and mental fatigue, especially compared to SSVEP (Xie et al., 2016). However, despite this advantage, its overall performance substantially lags behind that of SSVEP and c-VEP (Volosyak et al., 2020). Additionally, SSMVEP exhibits large inter-subject variability, with some users unable to achieve effective control, whereas c-VEP has shown to provide reliable and high performance for all users (Volosyak et al., 2020; Martínez-Cagigal et al., 2021; Thielen, 2025).

This study introduces the code-modulated motion visual evoked potential (c-MVEP), a novel paradigm that integrates the core principle of the reliable c-VEP protocol with the comfortable, low-fatigue characteristics of SSMVEP. In this paradigm, we propose modulating the stimuli with motion stimulation rather than flickers, following a pseudo-random sequence, to evoke a code-modulated response. While the motion-based stimulation improves user comfort (Xie et al., 2012, 2016), the implementation of the code-modulation may allow for more reliable control. The current study aims to investigate the feasibility and applicability of the c-MVEP paradigm in BCI applications.

To fully investigate the proposed c-MVEP paradigm, this study involves two experiments. First, an offline experiment to characterize and compare the neural responses of c-MVEP against c-VEP, SSMVEP, and SSVEP. Second, an online experiment to evaluate and compare the decoding performances of each of the paradigms in a BCI context.

2 Materials and Methods

2.1 Participants

A total of 28 subjects participated in this study. All subjects had normal or corrected-to-normal vision. Of all participants, only one participated in both offline and online experiments. The offline experiment included a total of 9 participants (2M, 7F, average age of 28.6 ± 4.0 , between 24-35)..

In the online experiment, 20 subjects participated. However, one participant was excluded due to task disengagement. A total of 19 participants (14M, 5F, average age of 23.7 ± 3.9 , between 19-35) were analyzed for the online experiment, of which **four participants reported having previous experience with a BCI system, consistent with recruitment via voluntary participation.**

Prior to the experiment, all participants received detailed information about the experimental procedure and any possible risks involved, all had the opportunity to opt out at any time during the experiment. Those who agreed to participate in the study signed a consent form and were financially compensated after their participation. The study was designed and conducted in accordance with the Helsinki Declaration and was approved by the ethical committee of the medical faculty of Duisburg-Essen University (24-11957-BO). All data were stored anonymously.

2.2 Hardware

Both the offline and online experiments were done in the same environment under comparable conditions using the same hardware.

The EEG data was acquired with a **g.USBamp amplifier (g.tec, Schiedlberg, Austria) using sixteen gel-based passive scalp electrodes placed according to the international 10-5 system at the following positions: $P_7, P_3, P_z, P_4, P_8, PO_7, PO_3, PO_z, PO_4, PO_8, O_1, O_z, O_2, O_9, I_z$, and O_{10} . The reference and ground electrodes were placed at C_z and AF_z , respectively. The EEG signals were recorded with a sampling frequency F_s of 600 Hz. Applying an abrasive electrode gel reduced the impedance to below 5 k Ω .**

The EEG data from the offline experiment were digitally filtered and epoched in software. **A notch filter was applied to remove 50 Hz power line noise. Then, the raw EEG data were bandpass filtered from 1.0 to 40.0 Hz using a fourth-order, zero-phase, infinite-impulse-response (IIR) Butterworth filter, baseline-corrected, and sliced into trials of the trial length. During the online experiment, the incoming continuous EEG data stream was causally filtered in real time, using a software-implemented, stateful fourth-order Butterworth bandpass filter (2.0 to 40.0 Hz).** To ensure consistency, the data used to train the classifier were processed with this same digital filtering procedure.

Both experiments were conducted on a Dell Precision desktop computer equipped with an Intel Core i9 processor (3.70 GHz) and an NVIDIA RTX 3070 graphics card, running Microsoft Windows 11 Education. Visual stimuli were presented on an Acer Nitro XV252Q F liquid-crystal display (1920 $\times$ 1080 pixels) with a refresh rate of 360 Hz.

We designed the experiments using the Unity game engine (Unity Technologies, San Francisco, CA, USA), version 6000.0.31f1.

2.3 Stimulus Generation

This study included four types of stimulation. Specifically, the newly proposed **c-MVEP with code-modulated motion stimulation, and its flickering counterpart, the code-modulated flickering c-VEP**. Both used the same pseudo-random sequence, a smoothed version (c-MVEP) and the original binary sequence (c-VEP). Additionally, we implemented the frequency-based motion stimulation of SSMVEP and its frequency-based flickering counterpart, SSVEP.

The motion conditions, c-MVEP and SSMVEP, used a **radial zooming of the stimulus, as this stimulation motion was shown to be preferred over Newton rings in SSMVEP** (Chai et al., 2019), and implemented effectively in Chai et al. (2020). The flickering of c-VEP and SSVEP was implemented as a binary flickering between black and white.

To implement the zooming in c-MVEP, we used a modified pseudo-random sequence, and the original binary pseudo-random sequence for flickering in c-VEP. **SSMVEP and SSVEP used distinct frequencies for stimulation**. Figure 1 shows a visual representation of the waveforms underlying the stimulation for each of the conditions. Figure 2 shows how these waveforms were translated into motion and flickering.

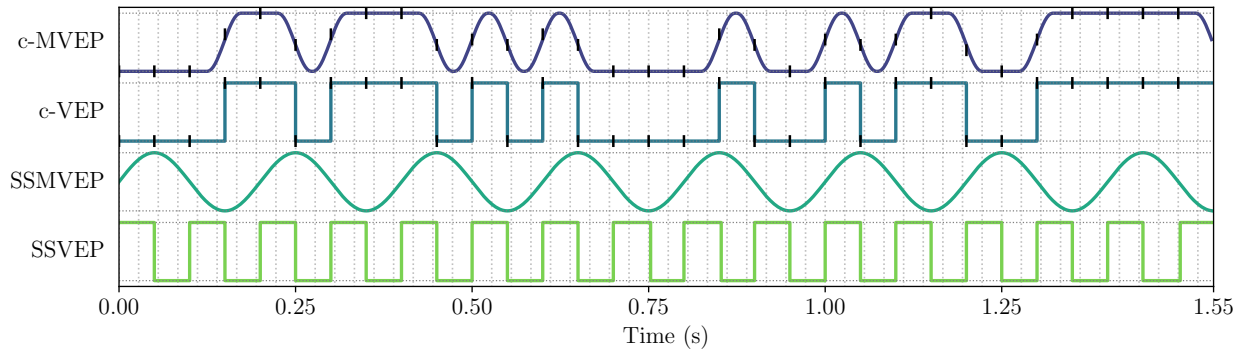


Figure 1: **Stimulation waveforms from the offline experiment.** The different waveforms used in the offline experiment for the four conditions: c-MVEP, c-VEP, SSMVEP, and SSVEP. The dotted vertical lines represent each 10th frame, at a monitor refresh rate of 360 Hz. **The black ticks in the c-MVEP and c-VEP conditions represent individual bits, of a presentation rate of 20 Hz, meaning 18 frames per bit.** SSMVEP has a flip-frequency of 10 Hz, meaning a sine wave of 5 Hz was used, SSVEP was presented at 10 Hz.

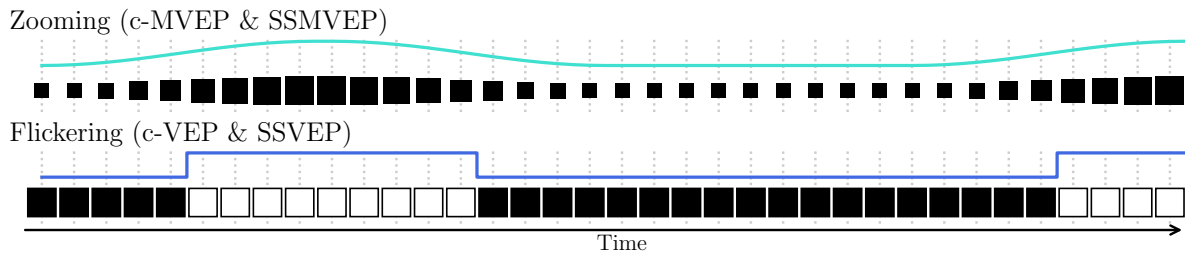


Figure 2: **Zooming and flickering stimulation.** Partial waveforms of the zooming conditions (c-MVEP and SSMVEP) and the corresponding stimulus size modulation. Similarly for the flickering conditions (c-VEP and SSVEP) a partial waveform is shown, together with the stimuli that are changing between black and white depending on the state of the waveform.

2.3.1 Code-Modulated Stimulation for c-MVEP and c-VEP

To visually stimulate the targets in a code-modulated-based BCI system, the stimuli flicker or move according to pseudo-random sequences. For c-VEP, this is often done using a binary maximum length sequence (m-sequence) (Thielen, 2025), which is generated by a linear feedback shift register (LFSR). An m-sequence has great autocorrelation properties, specifically, a correlation of 1 for no shift and $-1/K$ for all other shifts with K the length of the m-sequence (Golomb, 1982; Meel, 1999; Holmes, 2007). In c-VEP, this binary code represents the black ('0' in the code) and white ('1' in the code) flickering.

Here, we created the initial m-sequence c_0 with feedback taps at register positions 5 and 2, corresponding to the primitive polynomial of $p(x) = x^5 + x^2 + 1$. The initial register state was set to $r(0) = (1, 1, 1, 1, 1)$. This produced the following 31-bit m-sequence c_0 for the offline experiment:

$$c_0 = 0001101110101000010010110011111. \quad (1)$$

For the online experiment, we set four m-sequences such that the sequence started with a 1, creating a smooth start of the trial for the c-MVEP condition. Specifically, we right-circularly shifted the initial sequence c_0 by 1 bit, to create c_1 . The second sequence c_2 was a 3 bit right-circularly shifted version of c_0 . The third sequence c_3 shifted c_0 by 5 bits, and c_4 by 8 bits.

In their original form, these binary sequences were used to directly modulate the luminance of the stimuli, mapping the binary states to black ('0') and white ('1') for c-VEP. However, directly applying a binary (square-wave) modulation to movement parameters, specifically the zooming of a stimulus between predefined maximum and minimum sizes, would result in abrupt transitions. Such discontinuities could lead to perceptually rough motion and may even be experienced as flickering rather than a smooth movement. To address this, the binary m-sequences were transformed into continuous-valued sequences, with smooth transitions while preserving the original properties of an m-sequence. This is done by smoothing the edges of the m-sequence, similar to the approach used in a code-modulated auditory evoked potential (c-AEP) study (Scheppink et al., 2024). This manner of smoothing results in a sequence where the original duration of events and overall timing is retained, but the rising and falling edges are smoother, thereby improving the fluidity of the stimulation motion.

Specifically, we smoothed the edges using a raised sine function. First, the original m-sequence was upsampled by a factor of F_b , corresponding to the number of display frames per bit, here $F_b = 18$, derived from the monitor's refresh rate of 360 Hz and the 20 Hz presentation-rate of the m-sequence. Each transition between consecutive bits was identified. For each rising edge ($0 \rightarrow 1$) and falling edge ($1 \rightarrow 0$), a transition window of length $L = \alpha F_b$ was placed around each detected transition, where α denotes a smoothing factor. Here, $\alpha = 1.0$, resulting in a transition window equal to one bit duration. Within each transition window, indexed by $t = 0, \dots, L$, the sharp edge was replaced by a smooth sinusoidal function,

$$s(t) = \frac{1}{2} \left(1 + \sin \left(\pi \left(\frac{t}{L} - \frac{1}{2} \right) \right) \right), \quad t = 0, \dots, L \quad (2)$$

which smoothly connected 0 and 1. Rising edges were represented by $s(t)$, whereas the falling edges used the reversed function $1 - s(t)$.

The m-sequence was only modified within the transition windows, outside of these windows, the sequence remained unchanged. This created a smooth motion modulator, without introducing temporal artifacts or distorting the properties of the m-sequence, see Figure 1. The correlation properties of the m-sequence were minimally changed, as shown in Figure S1 of the Supplementary Material.

Although the resulting smoothed sequence ranged between 0 and 1, we did not allow motion in this full range. Rather, the stimulus size was limited to a minimum of 50 % and a maximum of 100 % of its original size, such that it zoomed between half and its full original size, rather than scaling down to zero, see Figure 2 and Figure 3.

A stimulus cycle duration is determined by the presentation rate (here set to 20 Hz) and the length of the sequence, yielding $31/20 = 1.55$ s for a cycle. Usually, a presentation rate of 60 Hz is used, together with a 63-bit code (Martínez-Cagigal et al., 2021). However, in order to maintain a zooming motion, we reduced the presentation rate, such that enough frames could be used to show the motion, and to stay close to the frequency range used in SSMVEP literature.

2.3.2 Frequency Stimulation for SSMVEP and SSVEP

In frequency-based BCI systems, such as SSVEP and SSMVEP, visual stimuli are modulated either through flicker or continuous motion according to constant, predefined frequencies. In the case of SSMVEP, it is most common to use a motion that includes a motion reversal, such as zooming in and zooming out. To achieve a perceptually smooth motion, sine-based stimulation waves are commonly used. Nevertheless, the use of square- or triangle-based stimulation waves has also been studied (Yan and Xu, 2020; Han et al., 2022; Kwon et al., 2022). Here, the SSMVEP stimulation is done with a sine-wave, using the following:

$$S(f, i) = \frac{1}{2}(1 + \sin(2\pi f(i/V_{RR}))) \quad (3)$$

where f is the stimulation frequency (Hz), i the frame index, and V_{RR} the screen’s vertical refresh rate. This is based on Nakanishi et al. (2014a); Chen et al. (2019) in which the approximation method for SSVEP is used, where a frequency is approximated with a variable number of frames in a period. We opted to use this strategy since it increases the number of frequencies that can be used with a specific refresh rate.

Similar to c-MVEP, we did not allow motion in this full range for SSMVEP. Again, the stimulus size was limited to a minimum of 50 % and a maximum of 100 % of the original stimulus size. For the offline study, a motion-inversion frequency of 10 Hz is chosen, specifically, $f = 5$ Hz. For the online study, with four targets, motion inversion frequencies, or flip-frequencies, of 9.0, 10.0, 11.25, 12.0 Hz are used.

For SSVEP, perceiving such flickering stimuli evokes a response with distinct oscillatory components at the fundamental flicker frequency and its harmonics (Vialatte et al., 2010). SSVEP stimulation in the range of 6 to 15 Hz was shown to evoke the strongest responses (Ajami et al., 2018; Nakanishi et al., 2014b; Stawicki et al., 2015, 2019). Here, to be close to the c-VEP, a stimulation of $f = 10$ Hz was chosen for the offline analysis, and 9.0, 10.0, 11.25, 12.0 Hz for the online study. The stimulation was done using a square-wave, and without phase differences.

2.4 Experimental Protocol

This study included two experiments. First, we did an offline experiment, where for each of the stimulus protocols, a single stimulus was presented on the screen to analyse the characteristics of the responses, see Figure 3. The second experiment was an online BCI selection application, where for each of the stimulus protocols, four targets were presented simultaneously on the screen to evaluate the decoding performance, see Figure 3.

2.4.1 Experiment 1: Offline Response Analysis

In the offline experiment, only one stimulus was presented in the center of the screen. This stimulus had a visual angle of $4.34^\circ \times 4.34^\circ$, at a viewing distance of 60 cm. We chose this design to obtain clean signals while avoiding potential interference from neighboring stimuli across the different experimental protocols. Subjects were asked to focus on this stimulus after a 1.5 s cueing period indicating the start of stimulation. After the green cue, the stimulus was reset to the original state (maximum size and black) for 1.0 s before the stimulation started.

For the c-MVEP and c-VEP conditions, stimulation lasted three times the cycle length, here $3 \times (31/20) = 4.65$ s. For SSMVEP and SSVEP, stimulation was 5.0 s. For all conditions, there was an inter-trial interval of 0.5 s, after which the next trial started.

To avoid order effects, the order of the conditions was randomized for the first run, and presented in reverse order the second run. The experiment was designed as follows: for each condition, 10 trials were presented, after which a questionnaire for this condition was filled in. Then, the next condition was started, where again 10 consecutive trials were presented, followed by the questionnaire. After all conditions had been observed and the final questions were answered, all conditions were presented again, in reversed order, each containing 10 trials, but did not include another questionnaire. This resulted in 20 trials per condition for each participant, see Figure 3.

Prior to starting the experiment, participants filled in a pre-experiment questionnaire, which asked about their previous BCI experience, current tiredness level, and the amount of sleep in the previous night. As mentioned above, after the first run of each of the conditions, the participants filled in a questionnaire to obtain the subjective rating. The questionnaire consisted of six questions that were answered on a 6-point Likert scale. The questions asked about the perceived visual discomfort caused by the stimulus, how easy it was to keep concentrating on the target, how disturbing the stimulation was, how often they experienced a loss of focus, how much they would like the stimulation in a BCI speller, and lastly, their overall rating of the stimulus.

2.4.2 Experiment 2: Online BCI Application

The online experiment explored the feasibility of using c-MVEP for BCI applications. All four stimulation paradigms were implemented to compare the newly proposed c-MVEP against the well-established protocols. In the experiments, the subjects were instructed to perform a copy-selection task. The numerical BCI system consisted of $N = 4$ stimuli with the same size as in the offline study, placed on a horizontal line on the screen with a spacing of 7.26° between stimuli. All stimulus conditions used the stimulation as described in subsection 2.3.

All conditions used the same classification approach, a template-matching canonical correlation analysis (CCA) classifier. The templates were built from user-specific data. To calibrate the classifier, the experiment started with a training block specific to the condition, to collect labeled training data. In this training block, the four stimuli were presented on the screen. Participants were instructed to focus on the cued stimulus until flickering or moving stopped. The cueing was done in green and lasted for 1.5 s, after which the stimulus was restored to its original color and size for 1.0 s, before flickering or moving started. For c-MVEP and c-VEP the stimulation lasted $3 \times (31/20) = 4.65$ s, for SSMVEP and SSVEP, the stimulation lasted 5.0 s. In total, each stimulus was attended to six times, for a total of 24 training trials per condition. After the training block of a condition, the participants were asked to fill out a questionnaire to rate their comfort on that condition.

After the questionnaire, the participants were introduced to the BCI with a 4-digit familiarization sequence to ensure they understood the selection of the digits. This was followed by the start of the testing block, which consisted of three runs, each with a 12-digit-long sequence. In each run, the full sequence to be entered was displayed at the top of the screen. To facilitate recognition of the next selection, the digit to be selected was highlighted in bold and blue within the sequence during gaze shifts and during stimulation, see Figure 3. The subjects had to shift their gaze to the corresponding target during a gaze-shifting period of 1.5 s.

Following the gaze-shift period, all $N = 4$ stimuli started flickering or moving simultaneously. When a classification was reached, the stimulation of all targets stopped, and the prediction was indicated using visual and auditory feedback and printed on the screen. A correct prediction produced a positive sound, and the stimulus was indicated in green for 0.5 s. For an incorrect prediction, the predicted target was indicated in red, together with a neural sound to provide clear auditory feedback without inducing negative reinforcement or user frustration. Incorrect selections could not be corrected, so participants had to continue to the next target. After a full sequence was selected, the participant could

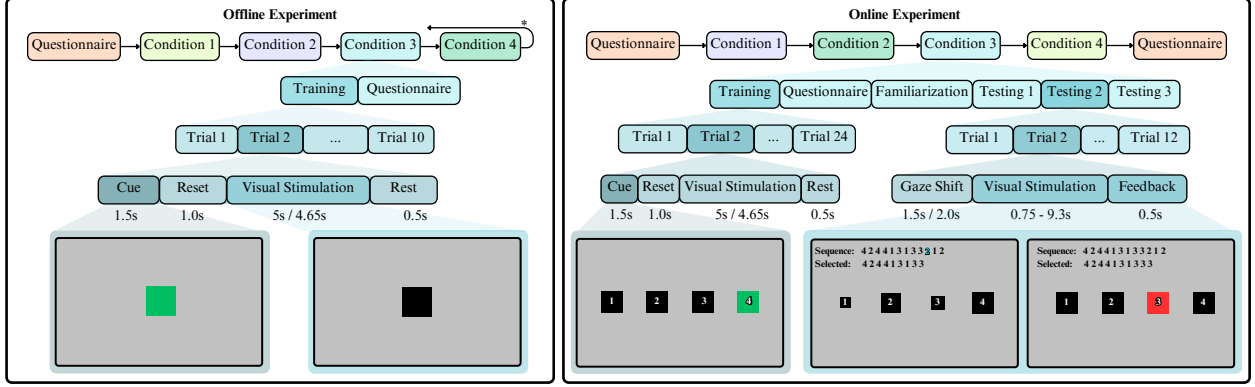


Figure 3: **Experimental Protocol.** A graphical representation of the experimental protocol of the offline (left) and online (right) experiments. In both experiments, all four conditions, c-MVEP, c-VEP, SSMVEP, and SSVEP were presented randomly. The visual stimulation duration was dependent on the stimulation paradigm, for c-MVEP and c-VEP this was 4.65 s, and 5 s for SSMVEP and SSVEP. The left figure shows the offline protocol, where one stimulus was shown on the screen. The left screen shows the cueing in green. The right shows the stimulation. Depending on the stimulation condition, this was either zooming (for c-MVEP and SSMVEP) or black-white flickering (c-VEP and SSVEP). Please note that, in the offline study, after the four conditions were presented in random order, they were presented again in reverse order, without the questionnaire. The figure on the right shows the online protocol, where four stimuli were presented simultaneously. From left to right, the screens depict the cueing in green for the training block, stimulation in the testing block, and the feedback for an incorrect selection.

continue to the next sequence at their own pace, see Figure 3. Between sequences, three participants indicated that the gaze-shifting period was too short. For those participants, the gaze-shift was increased to 2.0 s.

After each training block for a condition, participants completed a questionnaire to provide their subjective rating. This ensured that the condition’s BCI performance did not influence the answers. A total of seven questions were answered on a 6-point Likert scale. The questions inquired after their level of fatigue, how comfortable the stimulus was to look at, how easy it was to keep concentrating on the target, how disturbing the stimulation was, how often they experienced a loss of focus, how much they would like the stimulation in a BCI speller, and lastly, their general rating of the stimulus.

After the experiment, the participants filled in a post-experiment questionnaire. This inquired about the differences between zooming and flickering, specifically regarding which was easier to keep focus on, which caused the most eye strain, and what participants preferred. Lastly, they were also asked whether they observed a difference between the two flickering conditions and the two zooming conditions.

2.5 Data Processing and System Evaluation

2.5.1 Experiment 1: Offline Response Analysis

The data obtained in the offline experiment are examined in the time, frequency, and spatial domains. In the time-domain, we calculated the evoked waveforms at channel Oz to get an insight in the response to one stimulation cycle. A stimulation cycle was $31/20 = 1.55$ s for c-MVEP and c-VEP, and we used 1.0 s for SSMVEP and SSVEP. All trials are cut into non-overlapping segments of these cycles $\mathbf{X}_i \in \mathbb{R}^{C \times T}$, with C channels and T samples, resulting in 60 segments for c-MVEP and c-VEP and 100 segments for SSMVEP and SSVEP. Prior to computing the grand average for SSMVEP and SSVEP, each subject’s waveform was temporally shifted by up to ± 50 ms. This shift maximized the cross-correlation of the P1 peak with the group average, effectively correcting for inter-subject phase and latency differences.

To examine the frequency spectrum, the signal-to-noise ratio (SNR) at Oz was calculated for each subject and condition. The power spectral density (PSD) for each trial was estimated using a Hamming-windowed Fast Fourier Transform,

zero-padded to 10 seconds for a uniform 0.1 Hz frequency resolution. The resulting single-trial power spectra were averaged to reduce background noise and enhance the VEP spectral peaks. The SNR was calculated differently for the code-modulated and frequency-based paradigms to prevent spectral leakage, while allowing paradigm-specific spectral characteristics. For c-MVEP and c-VEP, we estimated the noise as the median power of 15 bins on each side, excluding the 2 adjacent bins. For SSMVEP and SSVEP, the noise was estimated from 20 neighbouring frequency bins on either side, excluding the 4 immediately adjacent bins to avoid signal leakage.

To investigate the spatial distribution of activation across the paradigms, we created topographic maps for the SNR. For each condition, the spatial analysis was limited to the primary target frequency driving the visual response, 20 Hz for c-MVEP and c-VEP, 5 Hz for SSMVEP, and 10 Hz for SSVEP.

2.5.2 Experiment 2: Online BCI Application

In c-VEP-, SSMVEP-, and SSVEP-based BCIs, canonical correlation analysis (CCA) has been successfully applied for template matching (Nakanishi et al., 2015; Bin et al., 2009) or frequency matching (Lin et al., 2007; Nakanishi et al., 2015). CCA is a multivariate statistical analysis method that reveals the underlying correlation between two sets of multidimensional variables (Hotelling, 1936). For two sets of signals $\mathbf{X}$ and $\mathbf{Y}$, the goal is to find linear projections $\mathbf{w}_x$ and $\mathbf{w}_y$ that maximise the correlation between $\mathbf{w}_x^\top \mathbf{X}$ and $\mathbf{w}_y^\top \mathbf{Y}$. Normally, $\mathbf{X}$ denotes the multi-channel EEG data, and $\mathbf{Y}$ the reference signals.

We adopted a unified decoding framework across all experimental conditions, using a template-based CCA approach. For the well-known c-VEP, SSVEP, and SSMVEP conditions, there exist methods specifically tailored to the paradigm that reach state-of-the-art performances, however, here we opt to use the same approach for each condition. This choice is motivated by the absence of evidence that any of the state-of-the-art methods from the c-VEP domain, such as reconvolution-based CCA (Thielen et al., 2015, 2021), generalizes effectively to the c-MVEP condition. Moreover, using a consistent decoding method across the conditions ensures that observed performance differences reflect the stimulation paradigm rather than the decoder.

Specifically, the CCA was implemented as a template-matching classifier, where it builds the templates as the averaged response to repeated stimulation of the same target stimulus:

$$\mathbf{T}_i = \frac{1}{R} \sum_r \mathbf{X}_r, \quad (4)$$

where $\mathbf{T}_i \in \mathbb{R}^{C \times M}$ and $\mathbf{X}_r \in \mathbb{R}^{C \times M}$ with $r = 1, \dots, R$ the number of single trials for the i -th class. Here, C denotes the number of channels, and M the number of samples.

Then, spatial filters are learned using CCA for the raw data in $\mathbf{X}$ and templates of $\mathbf{T}$. First, the templates $\mathbf{T}_i$ are stacked and repeated, according to the order of all $j = 1, \dots, J$ training trial in $\mathbf{X}$:

$$\mathbf{T} = [\mathbf{T}_{y_1}, \mathbf{T}_{y_2}, \dots, \mathbf{T}_{y_J}], \quad (5)$$

where $\mathbf{T} \in \mathbb{R}^{C \times M \cdot J}$, and $\mathbf{T}_{y_j}$ is the template of the class that is specified by the label y_j of the j th trial. All trials in $\mathbf{X}$ are concatenated to $\mathbf{S} \in \mathbb{R}^{C \times M \cdot J}$. Then, using $\mathbf{T}$ and $\mathbf{S}$, an optimized spatial filter $\mathbf{w}$ can be found using CCA:

$$\max_{\mathbf{w}, \mathbf{v}} \text{corr}(\mathbf{w}^\top \mathbf{S}, \mathbf{v}^\top \mathbf{T}) = \max_{\mathbf{w}, \mathbf{v}} \frac{\mathbf{w}^\top \mathbf{S} \mathbf{T}^\top \mathbf{v}}{\sqrt{\mathbf{w}^\top \mathbf{S} \mathbf{S}^\top \mathbf{w} \cdot \mathbf{v}^\top \mathbf{T} \mathbf{T}^\top \mathbf{v}}}. \quad (6)$$

During the online classification, restricting to the first CCA component only, spatial filters $\mathbf{w} \in \mathbb{R}^C$ and $\mathbf{v} \in \mathbb{R}^C$ can be used to spatially filter a single trial $\mathbf{X} \in \mathbb{R}^{C \times M}$ and multi-channel templates $\mathbf{T}_i$, respectively:

$$\mathbf{x} = \mathbf{w}^\top \mathbf{X}, \quad \mathbf{t}_i = \mathbf{v}^\top \mathbf{T}_i, \quad (7)$$

where $\mathbf{x} \in \mathbb{R}^M$ represents the weighted sum of the channels, and $\mathbf{t}_i \in \mathbb{R}^M$ the spatially filtered template response of the i th class. Then, the single trial is classified by selecting the template response $\mathbf{t}_i$ that maximizes Pearson's correlation:

$$\hat{y} = \arg \max_i \rho_i = \arg \max_i \frac{\mathbf{x}^\top \mathbf{t}_i}{\sqrt{\mathbf{x}^\top \mathbf{x} \cdot \mathbf{t}_i^\top \mathbf{t}_i}}. \quad (8)$$

Furthermore, we applied a dynamic stopping approach, similar to the margin method used in Thielen et al. (2015). Specifically, a classification result from the CCA was only accepted if a certain confidence threshold was met. A trial was accepted, and feedback was presented when the margin $\rho_m - \rho_r$, that is the difference between the maximum correlation ρ_m and runner-up ρ_r , was larger than or equal to 0.3. This threshold was set given pilot data, and was adjusted to 0.2 if, during the familiarisation sequence, it was too strict, leading to trial-time outs. The threshold was tested for every 0.05 s into the trial. If the end of a trial was reached and the threshold was not exceeded, no decision was forced.

The minimum window for classification was set to 0.75 s, the maximum time for classification was set to 9.3 s, which equals to six m-sequence cycles. When this maximum time was reached, and the confidence did not exceed the threshold, the classifier returned a no-decision, indicated by a '0', and the participant would continue with the next digit.

Furthermore, a combined expanding and sliding window approach was implemented. Initially, the window length was set to 0.75 s, after which the classifier attempted to decode the data. If the confidence threshold was not satisfied, the window was incrementally extended in steps of 0.05 s (equivalent to 30 samples). At each increment, the classifier re-evaluated the enlarged window, and this expansion continued until the threshold was met. If the window reached a maximum length of 4.65 s for c-MVEP and c-VEP, or 5.0 s for SSMVEP and SSVEP, without satisfying the confidence threshold, it was assumed that the initial portion of the window did not contain sufficiently informative data. Therefore, a sliding window approach was then used. The window was shifted forward in steps of 0.05 s, while maintaining the maximum window length (first-in-first-out). This process continued iteratively until the threshold was satisfied or the total maximum classification time of 9.3 s was reached.

In addition to accuracy, the online target selection performance was evaluated with the (practical) information transfer rate (ITR) (Wolpaw et al., 2002), measured in bits per minute,

$$ITR = \frac{\log_2 N + p \log_2 p + (1 - p) \log_2 \left(\frac{1-p}{N-1} \right)}{t/60}, \quad (9)$$

where N denotes the number of stimulus classes, p denotes the classification accuracy, and t denotes the average selection time (in seconds), including the inter-trial time of 2.0 s to 2.5 s. For more information on ITR and a publicly available tool of our lab, see our website ².

All statistical analyses were performed in Python using the SciPy and statsmodels libraries. First, Shapiro-Wilk tests were conducted to assess the normality of the data distributions for accuracy, ITR, and average selection time. As the data violated the assumption of normality, non-parametric tests were used for all analyses. That is, a Friedman test was used as an omnibus test to detect significant differences among the four conditions. Following the significant Friedman results, post-hoc pairwise comparisons were computed using Wilcoxon signed-rank tests. To correct for multiple comparisons, p-values were adjusted using the Bonferroni correction (significance threshold $\alpha_{adj} = .05/6 \approx .0083$).

²<https://bci-lab.hochschule-rhein-waal.de/en/itr.html>

3 Results

3.1 Experiment 1: Offline Response Analysis

3.1.1 Characteristics of Response

The response characteristics of the c-MVEP and c-VEP are shown in Figure 4. The signal response characteristics SSMVEP and the SSVEP conditions are shown in Figure 5. Please refer to the supplementary material for figures of the PSD spectrum and PSD spatial distribution. For all conditions, the grand-average evoked waveform at channel Oz, the SNR spectra at channel Oz, and spatial distributions were investigated.

The evoked waveform is a grand-average over all segments, trials, and subjects. For c-MVEP and c-VEP, each trial is cut into full m-sequence cycles, so $31/20 = 1.55$ s. For SSMVEP and SSVEP, each trial was cut into 1.0 s segments.

The evoked waveforms for c-MVEP and c-VEP did not show any frequency-specific oscillations, which is expected due to their code-modulated, non-periodic stimulation structure. This was supported by the SNR and PSD spectra, which did not show specific narrow-band peaks, but rather the energy was distributed more broadly across the frequency bands. The evoked waveform of c-MVEP showed a larger negative peak around 1.0 s, which is prominently visible, and did not occur in c-VEP. Its preceding positive peak did appear to be visible in both. The remaining waveform showed a similar pattern, although with clear, distinct differences.

The frequency spectra for c-MVEP and c-VEP showed a broad-band response to the m-sequence. For c-MVEP, it seemed that the broad-band response was more focused at the lower frequencies, whereas for c-VEP it also spread more beyond 10 Hz. In both, a peak at the 20 Hz presentation rate was present, with 2.11 dB for c-MVEP and 3.06 dB for c-VEP. Lastly, we inspected the spatial distributions of the 20 Hz activation. For c-MVEP, the activation was spread across multiple electrodes, with the main activation at Oz. For c-VEP, it appeared that Oz contained the majority of the 20 Hz activation, with less spread to neighboring electrodes.

The evoked waveforms for SSMVEP and SSVEP, aligned across participants prior to averaging, clearly showed sinusoidal characteristics. For SSVEP, the waveform predominantly reflected the stimulus' fundamental frequency, whereas SSMVEP had a more complex structure. This difference may partly be caused by an overall lower SNR observed for SSMVEP compared with SSVEP (SNR of 2.38 dB and mean peak-to-peak $5.69\mu V$ for SSMVEP, and SNR of 8.76 dB and mean peak-to-peak $12.34\mu V$ for SSVEP).

At the sinusoid frequency (5 Hz for SSMVEP, 10 Hz for SSVEP), SSMVEP elicited a peak SNR of approximately 2.36 dB. In contrast, SSVEP generated a peak SNR of approximately 8.76 dB. Furthermore, the SSMVEP response is characterised by the distinct harmonics at 10 Hz (SNR 2.01 dB) and at 15 Hz (SNR 1.15 dB). For SSVEP, the harmonics appeared stronger and more distinct at 20 Hz (SNR 10.54 dB) and 30 Hz (SNR 6.86 dB).

3.1.2 Subjective Rating

After the first run of a condition in the offline experiment, the participants were presented with a questionnaire that assessed their subjective rating of the stimulation. For each condition, six questions inquired about the visual comfort of the stimulus, ease of concentrating, disturbance, perceiving loss of focus, preference to use the system with more targets, and an overall score. Each question was answered on a 6-point Likert scale. Figure 6 shows the results of the nine subjects, the average and standard error values can be found in Table 1.

The results are ordered from highest to lowest rating, with higher mean scores indicating more favorable ratings for all measures. For eye-discomfort, c-VEP and SSMVEP obtained the highest, identical mean ratings, followed by SSVEP and c-MVEP. In terms of ease of concentration, c-VEP received the highest rating, followed by SSVEP, c-MVEP, and SSMVEP. Regarding the disturbance caused by the stimulation, SSMVEP was rated with the least disturbing scores, followed by c-VEP, SSVEP, and c-MVEP. For the loss of focus during stimulation, SSVEP received the highest ratings,

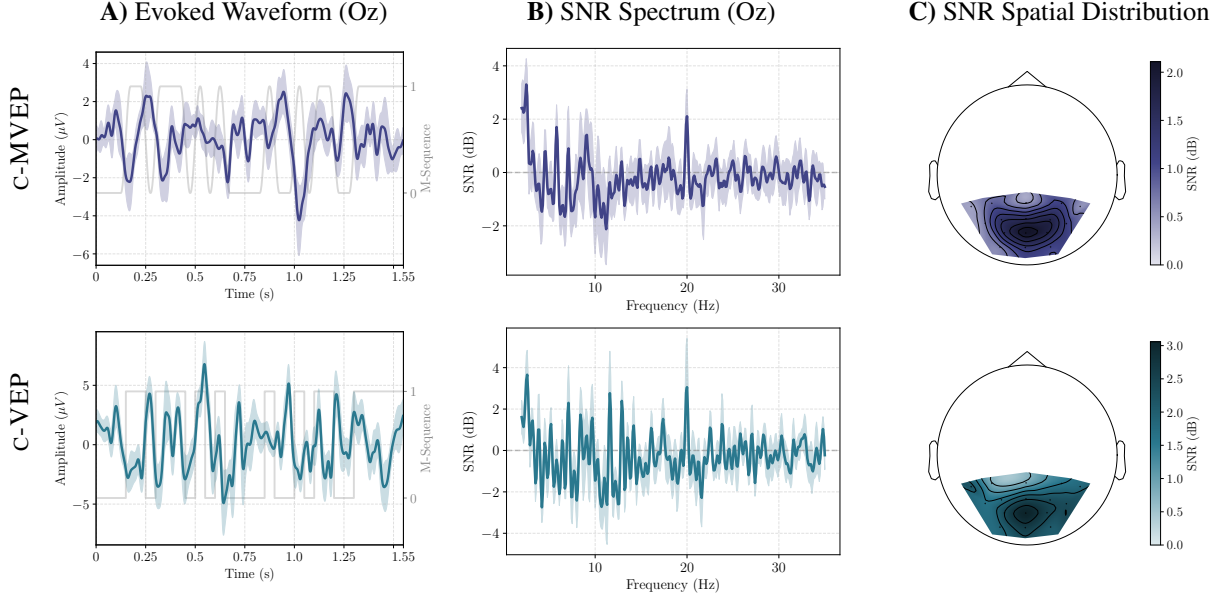


Figure 4: **Signal characteristics for c-MVEP and c-VEP.** The thick lines show the mean response, the shaded area the standard error. Columns show (A) the grand-average evoked waveform at channel Oz for a full cycle of 1.55 s. Here, the smoothened m-sequence (c-MVEP) and the binary m-sequence (c-VEP) are plotted in grey in the background. Column (B) shows the SNR spectrum at channel Oz. Column (C) shows the spatial distribution of the 20 Hz power.

Table 1: **Questionnaire Results Offline** Average (SE) results of the offline questionnaire. Higher scores indicate more favourable ratings.

	Comfort	Concentrating	Disturbance	Focus	Likability	Rating
c-MVEP	3.33 (0.44)	4.33 (0.47)	3.44 (0.41)	3.89 (0.45)	3.11 (0.39)	3.33 (0.33)
c-VEP	3.89 (0.42)	4.78 (0.28)	4.00 (0.44)	3.56 (0.53)	3.56 (0.56)	4.22 (0.40)
SSMVEP	3.89 (0.51)	4.22 (0.62)	4.22 (0.40)	4.00 (0.47)	3.00 (0.50)	3.11 (0.39)
SSVEP	3.67 (0.53)	4.44 (0.34)	3.67 (0.47)	4.22 (0.40)	3.67 (0.47)	3.78 (0.32)

followed by SSMVEP, c-MVEP, and c-VEP. Then, SSVEP received the highest scores for system likability, followed by c-VEP, c-MVEP, and SSMVEP. Lastly, c-VEP received the highest overall impression ratings, followed by SSVEP, c-MVEP, and SSMVEP.

3.2 Experiment 2: Online BCI Application

3.2.1 BCI Performance

The online selection task was performed for all four conditions c-MVEP, c-VEP, SSMVEP, and SSVEP. Using the four targets on the screen, the participants had to select numbers 1 to 4 according to a randomised 12-digit-long sequence. They fixated on the corresponding target, evoking a specific VEP response. Using CCA, this response was decoded, and feedback was given to the participant. The results in the online selection tasks are presented in Figure 7. Table 2 denotes the mean performance and standard error for each condition and performance metric.

For the accuracy, a Friedman test revealed a significant main effect of condition ($\chi^2 = 34.59$, $p < .001$). Mean accuracies were highest for c-VEP ($97.81 \pm 3.31\%$), followed by SSVEP ($93.42 \pm 2.11\%$), c-MVEP ($85.67 \pm 3.31\%$), and SSMVEP ($64.91 \pm 5.32\%$) (mean $\pm$ SE). Post-hoc pairwise comparisons were conducted using a one-sided Wilcoxon signed-rank test, with a Bonferroni correction ($\alpha_{adj} = .0083$). We found that c-MVEP reached significantly higher accuracies than SSMVEP ($p < .001$). SSVEP significantly outperformed SSMVEP and c-MVEP (both

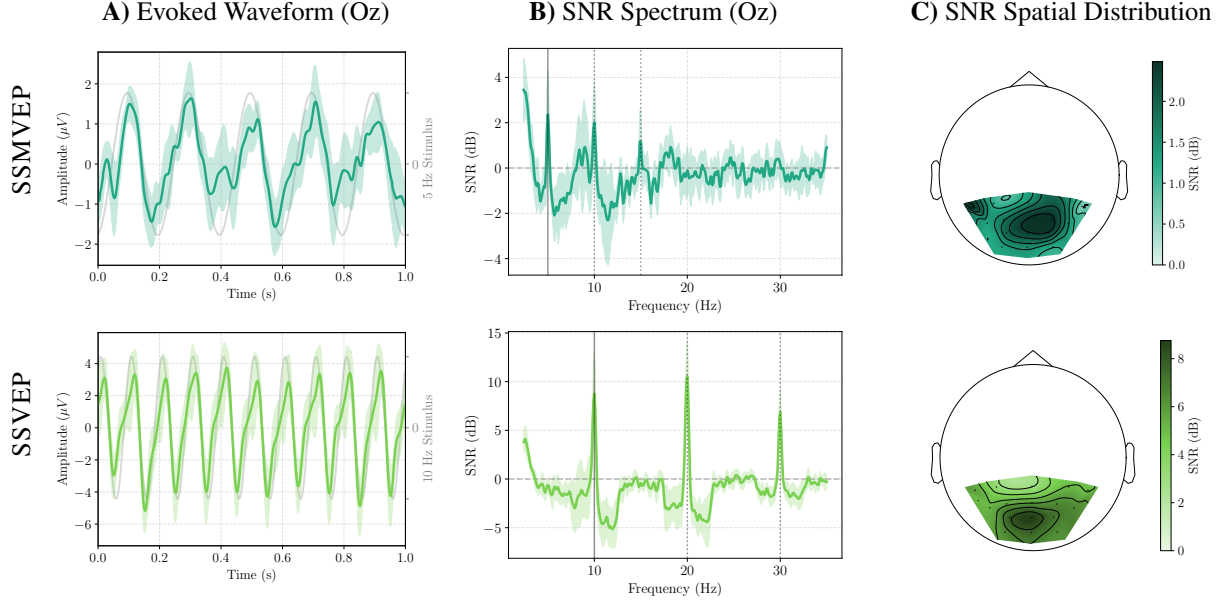


Figure 5: **Signal characteristics for SSMVEP and SSVEP.** The thick lines show the mean response, the shaded area the standard error. Columns show (A) the grand-average evoked waveform at channel Oz for 1.0 s, with the original stimulation waveform in light gray. Column (B) shows the SNR spectrum at channel Oz. Column (C) shows the spatial distribution of the 5 Hz (SSMVEP) and 10 Hz (SSVEP) power.

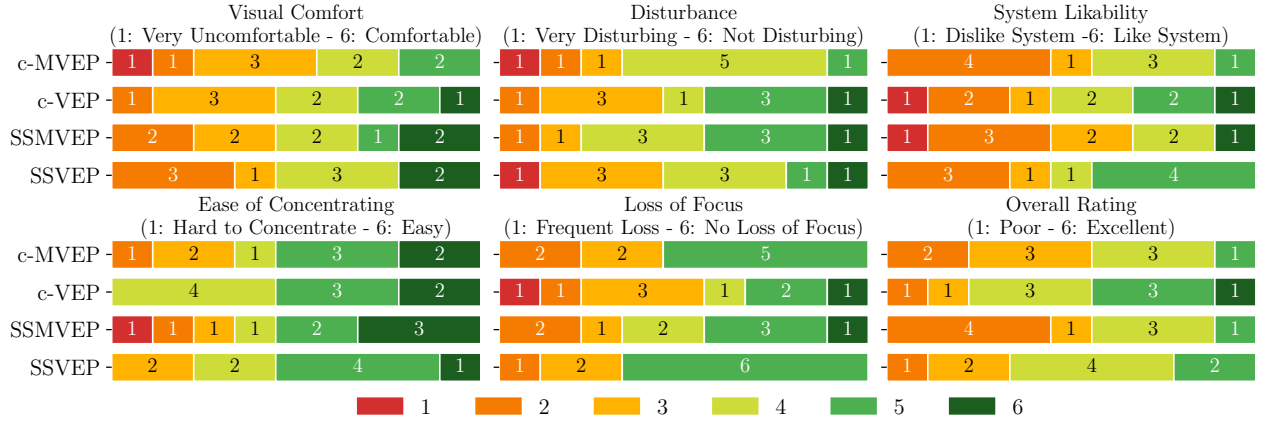


Figure 6: **Results Offline Questionnaire.** The answers to the questionnaire from the offline experiment. Questions were answered on a 6-point Likert scale. Here, 1 indicates a more negative rating, and 6 a more positive rating. The numbers in the bars indicate the number of participants giving that specific score.

$p \leq .002$). Additionally, c-VEP outperformed SSMVEP and c-MVEP (both $p \leq .001$). Lastly, there was no significant difference between SSVEP and c-VEP ($p = .018$).

The selection time also differed significantly between conditions ($\chi^2 = 38.56$, $p < .001$). c-VEP had the shortest mean selection times (1.15 ± 0.08 s), followed by SSVEP (1.94 ± 0.28 s), c-MVEP (2.61 ± 0.25 s), and SSMVEP (4.18 ± 0.37 s) (mean $\pm$ SE). Post-hoc Wilcoxon signed-rank tests with Bonferroni correction revealed that c-MVEP reached significantly faster selection times than SSMVEP ($p < .001$), but was outperformed by SSVEP ($p = .027$). SSVEP furthermore reached significantly lower selection times than SSMVEP ($p < .001$). Lastly, c-VEP significantly outperformed c-MVEP, SSMVEP, and SSVEP (all $p \leq .001$).

ITR results showed a significant difference between conditions ($\chi^2 = 41.53$, $p < .001$). Average ITR values were highest for c-VEP (35.56 ± 1.36 bits/min), followed by SSVEP (28.12 ± 2.24 bits/min), c-MVEP (20.11 ± 2.46 bits/min), and SSMVEP (10.13 ± 2.41 bits/min) (mean $\pm$ SE). The post-hoc Wilcoxon signed-rank tests with Bonferroni correction revealed that c-MVEP significantly outperformed SSMVEP ($p < .001$), but was significantly outperformed by c-VEP and SSVEP ($p < .001$). Both SSVEP and c-VEP also reached significantly higher ITR than SSMVEP ($p < .001$), and c-VEP significantly outperformed SSVEP ($p < .001$).

To assess whether the effect of adding motion differed between SSVEP and c-VEP paradigms, a subject-wise motion effect score (motion - flicker) was computed for each paradigm. These difference scores were compared using a two-sided Wilcoxon signed-rank test.

Adding motion reduced accuracy by 28.51 ± 5.51 % in SSVEP paradigm, compared to 12.13 ± 3.46 % in the c-VEP paradigm. The reduction in accuracy due to adding motion was significantly different between the paradigms ($p = 0.031$). Selection time increased by 2.24 ± 0.43 s in the SSVEP paradigm when motion was added, compared to 1.46 ± 0.24 s in the c-VEP paradigm; however, this difference was not statistically significant ($p = .060$). Similarly, the ITR decreased by 17.99 ± 2.49 bits/min in the SSVEP paradigm and by 15.45 ± 2.42 bits/min in the c-VEP paradigm, with no significant difference in motion-related reduction between the paradigms ($p = .623$).

During the online task, the maximal trial duration was reached at least once in all conditions. We did not force a classification when the maximum trial duration was reached. Nevertheless, the classifier may still have been able to make a prediction, albeit at a lower confidence. To assess potential classifier behaviour under these circumstances, we performed a post-hoc analysis on the timed-out trials using the classifier state saved during the experiment. To stay consistent with the original approach, classification was based on the maximum window length extracted from the end of each timed-out trial.

For c-MVEP, 7.9 % of all trials across participants reached the time limit. If classifications had been forced, an average accuracy of 87.0 % would have been reached. For c-VEP, only a single trial reached the maximum duration and was incorrectly classified in the post-hoc analysis, leaving the overall mean decoding accuracy unchanged at 97.8 %. For SSMVEP, 17.5 % of trials reached the maximum duration. Forced classification of these trials would have resulted in a mean accuracy of 69.9 % could have been reached. Lastly, for SSVEP, 2.2 % of trials timed-out, and forcing a classification would have resulted in a mean accuracy of 94.7 %.

Table 2: **Average Performance and SD.** Mean and standard error values for all metrics (Accuracy, Selection Time, ITR) for each condition (c-MVEP, c-VEP, SSMVEP, SSVEP).

Condition	Accuracy	Selection Time	ITR
c-MVEP	85.67 ± 3.31	2.61 ± 0.25	20.11 ± 2.46
c-VEP	97.81 ± 0.84	1.15 ± 0.08	35.56 ± 1.36
SSMVEP	64.91 ± 5.32	4.18 ± 0.37	10.13 ± 2.41
SSVEP	93.42 ± 2.11	1.94 ± 0.28	28.12 ± 2.24

3.2.2 Subjective Rating

In the online experiment, the participants completed a questionnaire after each training session, prior to starting the testing session of a condition, to avoid the performance of the BCI influencing the answers. It consisted of seven questions assessing the participant’s perceived fatigue level, visual comfort of the stimulus, ease of concentration, disturbance, perceived loss of focus, system likability, and an overall rating. After completing all experimental conditions, participants were asked to compare the flickering and zooming stimulation types by indicating which was easiest to maintain focus on, which caused the most eye strain, and their final overall preference. The answers are shown in Figure 8, with the average and standard error values in Table 1.

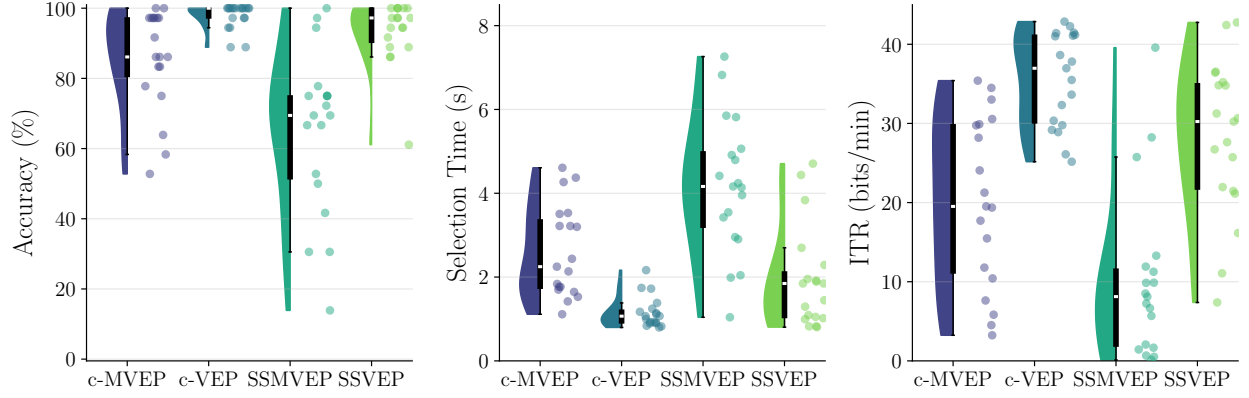


Figure 7: **Accuracy, Selection Time, and ITR Results for the Online Experiment.** Raincloud plots with the performances of the online BCI selection experiment for each condition (c-MVEP in dark-blue, c-VEP in teal, SSMVEP in green, SSVEP in lime green) per performance metric (Accuracy, Selection Time, ITR). Specifically, the left graph shows the distribution of the decoding accuracies for all participants and 12-digit sequences. The middle graph depicts the average selection time, excluding ITI time. The graph on the right shows the average ITR, which included ITI time. The half-split violin plot represents the density distribution of the data, the standard boxplot indicates the first and third quartiles, with the white horizontal lines denoting the median, and the jittered scatter points represent the averaged metric for each participant.

Table 3: **Questionnaire Results Online** Average (SE) results of the online questionnaire. Higher scores indicate more favourable ratings.

	Fatigue	Comfort	Concentrating	Disturbance	Focus	Likability	Rating
c-MVEP	4.26 (0.29)	4.11 (0.33)	4.32 (0.30)	4.58 (0.29)	4.11 (0.30)	4.05 (0.38)	4.79 (0.31)
c-VEP	4.26 (0.35)	3.79 (0.31)	4.37 (0.31)	4.47 (0.35)	4.05 (0.31)	4.05 (0.39)	4.74 (0.29)
SSMVEP	4.21 (0.29)	4.21 (0.26)	4.58 (0.30)	4.89 (0.35)	4.21 (0.29)	4.37 (0.38)	5.00 (0.28)
SSVEP	4.00 (0.32)	3.42 (0.30)	3.95 (0.32)	3.84 (0.32)	4.05 (0.30)	4.00 (0.43)	4.63 (0.34)

The non-parametric Friedman test was used to test for a significant difference. No significant difference was found for the reported fatigue, discomfort rating, ease of concentration, loss of focus, system likability, and overall preference (all $p \geq .073$). A significant difference was found for the disturbance ($\chi^2 = 8.475$, $p = .037$); however, the post-hoc Wilcoxon signed-rank test with Bonferroni correction (adjusted $\alpha = .0083$) revealed no significant differences between specific pairs of conditions (all $p \geq .020$).

Responses to the final questionnaire were analyzed separately, as these questions did not correspond to the individual conditions. These were binary yes/no questions and forced-choice preference questions (zooming or flickering). They were analyzed using two-sided binomial tests against the chance level ($p = .5$). Statistical significance was at $\alpha = 0.05$. A large majority of the participants (17/19) reported perceiving differences in the flickering stimuli (c-VEP and SSVEP), which was significantly above chance level ($p < .001$). In contrast, although most participants (14/19) reported perceiving differences in the zooming stimuli (c-MVEP and SSMVEP), but this did not reach statistical significance ($p = .064$). Lastly, no significant differences were found in the forced-choice preference questions regarding which was easiest to focus on (8 zooming, 11 flickering; $p = .648$), which caused the most eye strain (5 zooming, 14 flickering; $p = .064$), and for overall preference (9 zooming, 10 flickering; $p > .999$).

4 Discussion

This study proposed a novel code-modulated motion VEP paradigm and systematically compared it against each of its directly related protocols, including c-VEP, SSMVEP, and SSVEP. We proposed the c-MVEP paradigm as a moving

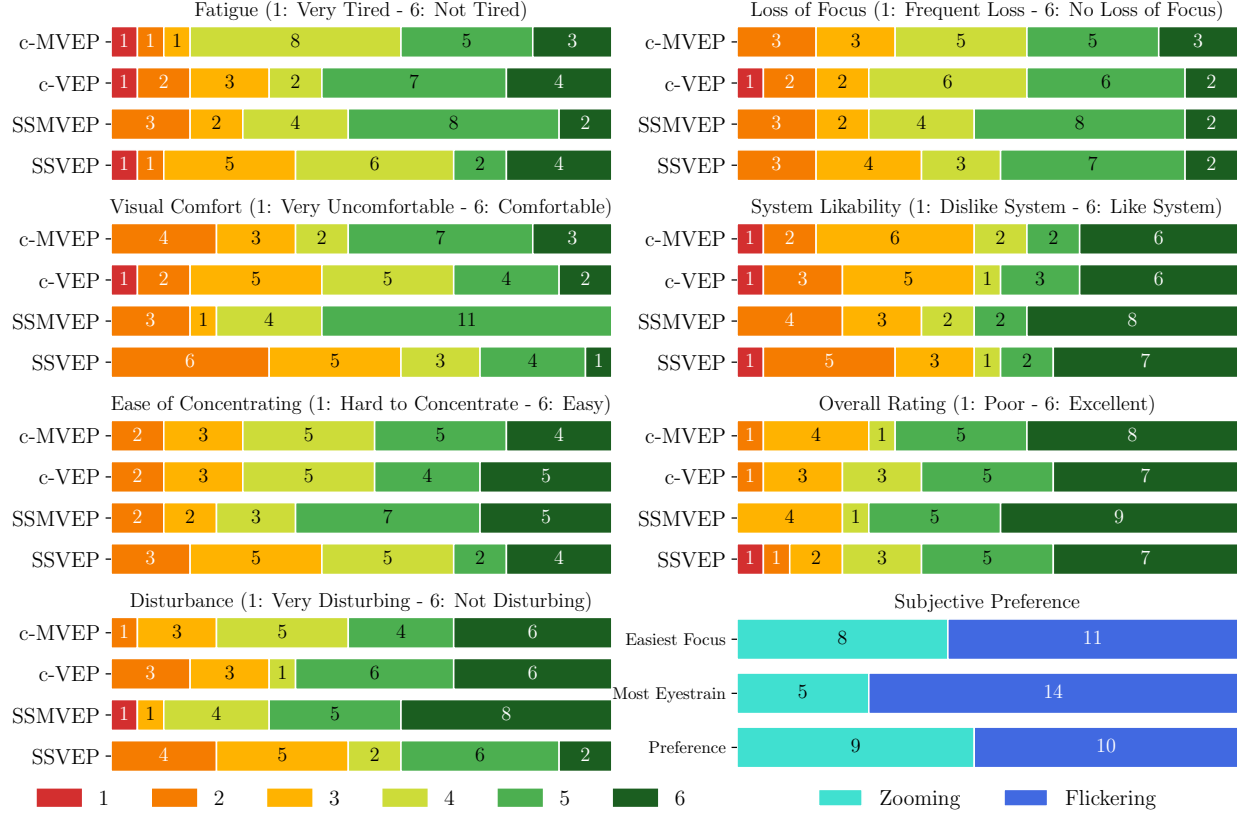


Figure 8: **Results Online Questionnaire.** The answers to the questionnaire from the online experiment. Questions 1 - 7 were answered on a 6-point Likert scale. Here, 1 indicates a more negative rating, and 6 a more positive rating. The bottom-right bars depict the answers to the binary forced-choice preference questions (zooming or flickering). The numbers in the bars indicate the number of participants giving that specific score.

variant of the well-performing and robust c-VEP stimulation protocol, as a more comfortable yet reliable alternative. The study consisted of both an offline and online experiment to investigate the response characteristics of c-MVEP and to study the feasibility of the c-MVEP for BCI applications and the online BCI performance across all four paradigms.

4.1 Experiment 1: Offline Response Analysis

The offline experiment was used to investigate the response characteristics of c-MVEP compared to its three related paradigms, using an individual stimulus on the screen. For c-VEP and c-MVEP, the grand-averaged evoked waveform (Figure 4) is non-trivial to interpret due to the core principles of the code-modulated paradigm, however, we may see that the motion caused different responses than the c-VEP flickering, although certain patterns might be recognized in both.

In the frequency spectra, we observed that the c-MVEP evoked a broad-band response, similar to c-VEP, although more focused in the lower part of the frequency domain. In line with Bin et al. (2009), the responses evoked by c-VEP and c-MVEP contain multiple prominent frequencies, spread throughout the spectrum, confirming that both are evoking the expected broad-band response. Striking in the c-MVEP plot, however, is that there appeared to be more variability between subjects than could be observed in c-VEP.

The slightly lower amplitudes of the evoked waveform in c-MVEP and the overall larger inter-subject variability for both the time and frequency domain could be explained by the fact that some participants noted that it was hard to concentrate on the zooming stimuli, which may have influenced the response.

For the spatial distribution, we observed that c-MVEP appeared to be more spread around and away from the Oz electrode than c-VEP, which was focused around Oz. This is in line with what was observed in the SSMVEP domain, where SSVEP was mainly focused in the occipital region, and SSMVEP spread out more towards the middle temporal vision region (Han et al., 2018b). Nonetheless, our analysis was limited to the distribution of the 20 Hz power, equal to the presentation rate of the m-sequence. Alternatively, the c-MVEP motion and c-VEP flicker responses could be characterized using all frequency components associated with the various event durations within the m-sequence.

The evoked waveform from SSVEP and SSMVEP (Figure 5) showed a clear sinusoidal response, which is expected and similar to SSMVEP literature (Han et al., 2018a). For SSMVEP, it appeared that the waveform had higher variability in subjects and did not synchronize at the stimulation frequency as SSVEP appeared to do. Both SSMVEP and SSVEP showed strong SNRs for the fundamental frequency and the harmonic frequencies. The harmonic responses were stronger for SSVEP than for SSMVEP, which is not in line with Stawicki and Volosyak (2020), who observed high harmonic responses for SSMVEP, nor in line with Han et al. (2018a), who reported only less obvious spectrum peaks for the second harmonic. Similar to what we observed for c-MVEP and c-VEP, here SSMVEP showed activity more spread away from Oz than SSVEP, where the activity was focused at Oz, in line with Han et al. (2018b).

From the SNR plots, it may be observed that adding motion to the SSVEP paradigm seems to result in a larger decrease in SNR than when adding motion to the c-VEP paradigm.

4.2 Experiment 2: BCI Performance

The online experiment investigated the feasibility of using c-MVEP for online BCI applications and compared the protocol to its related stimulation paradigms. Overall, both flickering paradigms outperformed the motion-based paradigms, for accuracy, selection time, and ITR. Specifically, c-VEP was the best performing protocol, and reached significantly higher accuracies than c-MVEP and SSMVEP. c-VEP furthermore had the shortest selection times, significantly shorter than c-MVEP, SSMVEP, and SSVEP. Finally, c-VEP also reached the highest ITR, which was significantly higher than c-MVEP, SSMVEP, and SSVEP.

Between c-MVEP and SSMVEP, it was found that c-MVEP reached significantly higher accuracies (c-MVEP: 85.67%, SSMVEP: 64.91). It has to be noted here that the SSMVEP performance observed in the current study did not correspond to performances reported in Chai et al. (2019, 2020); Volosyak et al. (2020), where accuracies of 89.7 %, 94.9 %, and 91.1 %, respectively. However, the current study opted for CCA with user-specific templates, whereas in the aforementioned studies, CCA with sine and cosine templates was used. Nonetheless, one would argue that the former would be better suited for the participant than the latter. In addition, SSMVEP frequencies used in the current study were chosen to be close to the c-MVEP movement, and are also in line with previous studies (Chai et al., 2019, 2020; Gao et al., 2019a,b), which again reported higher accuracies than observed in the current study.

For selection times, a similar finding is observed, where c-MVEP (2.61 ± 1.09 s) was significantly faster than SSMVEP (4.18 ± 1.62 s). Similarly, c-MVEP reached significantly higher ITR than SSMVEP. When comparing this to c-VEP and SSVEP, a similar pattern is observed, where c-VEP was significantly faster in terms of selection time and reached significantly higher ITR. Only for accuracy, there was no significant difference.

Lastly, in the plots in Figure 7, there appears to be more inter-subject variability in subjects for SSMVEP than c-MVEP. Similarly, SSVEP shows more inter-subject variability than c-VEP. This would indicate that the code-modulated stimulation paradigm more consistently reaches higher accuracies, faster selection times, and ITR than the frequency-based stimulation paradigm. Additionally, this might indicate that the frequency-based protocols rely more on optimization for individuals than the code-modulated protocols.

When inspecting the effect of adding motion to SSVEP and c-VEP paradigms, it was observed that for the decoding accuracy, the reduction in accuracy and increase in selection times when going from SSVEP to SSMVEP was significantly larger than when going from c-VEP to c-MVEP. No such difference was observed for ITR. For both paradigms, it could be observed that, when adding motion, there is more subject variability. Nonetheless, the code-modulated paradigm seems to be more robust against adding motion into the paradigm than the frequency paradigm.

4.3 Subjective Rating

To assess user experience, subjective ratings were collected for all conditions in both the offline and online experiments. For the offline experiment, participants answered the questionnaire following the first run of a given condition, whereas in the online experiment, the questions were presented after the training block but prior to the testing block. The questions were used to assess the level of comfort of the different stimulation conditions. From both the questionnaires, we found no preference for SSMVEP over SSVEP for comfort, which is not in line with results from subjective analysis in Chai et al. (2019). However, the results were in line with Volosyak et al. (2020), where no statistical difference was found for comfort ratings between c-VEP, SSVEP and SSMVEP. Similarly, we found no preference for c-MVEP over the flickering c-VEP and SSVEP conditions.

To assess subjective preference between the moving and flickering paradigms, participants completed a final questionnaire at the end of the online experiment, after experiencing all conditions. No significant differences were found between zooming and flickering for ease of focusing, causing most eye-strain or overall preference. This study showed no clear benefit on subjective comfort for any of the conditions.

This is striking, since in the study we used a presentation rate of 20 Hz for c-MVEP and c-VEP, instead of the commonly used 60 Hz, and it was shown that c-VEP flickering at higher presentation rates was perceived as less fatiguing (Martínez-Cagigal et al., 2021). Since the current study used a lower presentation rate, it would be expected that the motion stimulation was preferred over this flickering.

4.4 Limitations and Future Work

The current findings show the feasibility of adding motion to the c-VEP protocol for BCIs. Nonetheless, further research is necessary to optimize the stimulation and decoding paradigm.

First, the current performances were found with a basic template-matching classifier, which was not optimized for any of the current protocols, but was still able to evoke high BCI performance. Future work should focus on improving the decoding by taking inspiration from decoding approaches of the c-VEP domain, such as the reconvolution-based CCA (Thielen et al., 2021), as well as the SSMVEP domain, such as task-related component analysis (Nakanishi et al., 2018; Stawicki et al., 2021). Especially when using the c-VEP optimized decoding approaches, we should investigate how to further adapt and optimize the analysis to suit the motion and the smoothed m-sequence.

Second, it appears that c-MVEP is a better alternative for high-contrast flickering than SSMVEP, especially for those who prefer the motion over the flickering stimuli. However, the current implementation of motion could be improved by using a less invasive motion, for instance, by moving textures on the stimulus instead of the full stimulus itself. Furthermore, the amplitude of the motion could be optimized. Specifically, here we opted for a 50 % reduction in size, however, a lower zooming amplitude could potentially lead to more visual comfort.

Additionally, a lower zooming amplitude could alleviate the disadvantage of the currently used implementation. Specifically, the size of the stimulus matters for ERP response, creating a disadvantage for c-MVEP and SSMVEP compared to c-VEP and SSVEP, in the current study, since the latter two were always presented at 100 % size. Additionally, when opting for a size change, we would advise having the center-point correspond to the size of the flickering stimuli, such that a 50 % zoom corresponds to zooming between 75% and 125% to make the size component more fair.

A benefit from the motion stimuli is that, as shown by Gao et al. (2021a,b), it is less affected by competing stimuli than the flickering stimuli. This could be a major advantage of c-MVEP as an alternative to c-VEP, compared to other alternatives proposed, such as the non-binary grey-flickering c-VEP (Martinez-Cagigal et al., 2023).

After having improved further upon the stimulation movement used and optimizing the decoding paradigm for c-MVEP stimulation, the newly proposed c-MVEP may be a good and reliable alternative to c-VEP, SSMVEP, and SSVEP.

5 Conclusion

This study presented, for the first time, the newly proposed code-modulated motion visual evoked potential (c-MVEP). With the new visual stimulus protocol, we attempted to combine the high performance of c-VEP with the improved comfort of SSMVEP. We performed a systematic comparison between c-MVEP, c-VEP, SSMVEP, and SSVEP, and found that c-MVEP outperformed SSMVEP in terms of accuracy, selection time, and ITR. However, both c-VEP and SSVEP outperformed c-MVEP. There was no clear user preference for any of the conditions, that is, all conditions reached a similar comfort level. In summary, we proposed a novel visual stimulus protocol, which could be tailored to individuals and provide a competitive alternative when others do not work or feel less comfortable for specific individuals. This study has taken the first step towards a flicker-free c-VEP BCI, showing the potential for creating comfortable and reliable c-MVEP BCI applications.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

HS: Conceptualization, Methodology, Data Curation, Investigation, Formal Analysis, Visualization, Validation, Writing - Original Draft, Writing - Review & Editing; RH: Writing - Review & Editing; JT: Conceptualization, Methodology, Writing - Review & Editing; IV: Writing - Review & Editing, Resources, Project Administration, Funding Acquisition, Supervision.

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Supplemental Data

Data Availability Statement

The datasets analyzed in this study are not currently available for public access. All EEG recordings were anonymized; however, our existing ethical approval does not cover public data sharing.

Ethics statement

The studies involving humans were approved by the Ethics Committee of the Medical Faculty of the University of Duisburg-Essen, reference 24-11957-BO. All study procedures complied with applicable local legislation and institutional guidelines.

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